

Azim Ilkal

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Education

University of California, Davis

Davis, CA

B.S. Mechanical and Aerospace Engineering · GPA: 3.6

Expected June 2028

- Relevant Coursework: Fluid Mechanics, Dynamics, MATLAB, Mechanics of Materials, Electronic Systems.
- Activities: National Association of Rocketry, Aggie Propulsion & Rocketry Lab, Formula SAE, Run Club, IM soccer.

Experience

Birdstop

March 2025 – Present

Flight Systems Intern (Jun 2026 - Aug 2026)

Detroit, MI

- Conducted CFD wind-load analysis (ANSYS Fluent) of a quadcopter across a sweep of wind angles, deriving force/moment-balance equations for sliding, tipping, and lift-off failure modes to compute critical wind speeds.
- Identified the worst case velocity (35 mph) and validated Reynolds independence across a 9–54 m/s range.
- Designed and tested a perch shield that reduced drone drag up to ~47% and raised worst case velocity to 88 mph.

Field Engineer (Mar 2025 - Present)

Concord, CA

- Designed 50+ mechanical components in SolidWorks to be 3D printed for use in beacons and drones enabling tighter packaging and easier transportation.
- Reduced beacon manufacturing time by 50% by optimizing manufacturing and production processes.
- Deployed and validated 8 beacon systems across Spain, enabling air-traffic monitoring for Spanish ATC.
- Conducted 10+ live UAV flight tests to test battery time and calibrate flight settings for optimal control.

Unitrans

September 2025 – Present

Shop Assistant (Bus Technician)

Davis, CA

- Diagnosed and repaired mechanical, hydraulic, pneumatic, and electrical faults across a 250+ vehicle transit fleet.
- Performed preventative maintenance and safety inspections, documenting all work to DOT standards.

National Association of Rocketry

October 2025 – Present

Manufacturing Officer

Davis, CA

- Design and manufacture flight hardware for competition rockets, including nose cones, fins, and motor mounts.
- Fabricate components using machining and 3D printing, then integrate and test assemblies prior to launches.
- Launched to 12,106 ft and placed 2nd of 17 teams, ahead of UCLA, Yale, and UCF, at a Mojave Desert competition.

Formula SAE

January 2025 – December 2025

Composites Technician

Davis, CA

- Fabricated aerodynamic components (front/rear wing, and endplate) from carbon fiber for the competition vehicle.
- Machined molds on a CNC router and laser cutter to perform carbon-fiber layups.
- Aero package improved downforce and reduced drag, leading to a 3.5-second improvement in lap times.

Projects

Weather Station

December 2025 – February 2026

- Built a Raspberry Pi-based weather station with a 3D-printed housing integrating 5 sensors: a BME280 (temperature, pressure, humidity), a Hall-effect anemometer, a magnetic encoder for wind direction, and UV and rain sensors.
- Programmed the Pi in Python to read, process, and log all sensor data, validating readings against local meteorological reference data.

High Power Rocket

February 2025 – March 2025

- Designed, and launched a rocket to 3,000 ft, recovering it via parachute deployment, and earning L1 certification.
- Reinforced fins with fiberglass and bonded the airframe with epoxy to withstand high-thrust launch loads.

Skills

CAD & Analysis: Fusion 360, SolidWorks, CATIA, Onshape, Ansys Fluent, Ansys Mechanical, OpenRocket.

Manufacturing: 3D Printers (FDM & Resin), Laser Cutter, Band Saw, Bridgeport Mill, Soldering, CNC.

Programming and Data: Python, MATLAB, Arduino, Java, Excel.

Certifications: FAA Part 107, NAR HPR Level 1.